SOMAIYA
VIDYAVIHAR UNIVERSITY
Somaiya School of Basic and Applied Sciences

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Examination Seat No: 31011224039

CERTIFICATE

Somaiya Vidyavihar University
Somaiya School of Basic and Applied Sciences

BCA Sem-III
2025-2026

This is to certify that Mr/Ms. Murali Krishna M of BCA Sem-III Seat no 31011224039 has successfully completed Mini Project titled A Study on Personal Finance Habits in the subject of ELEMENTS OF STATISTICS LAB during academic year 2025-2026 as specified by the SOMAIYA VIDYAVIHAR UNIVERSITY.

Faculty in-charge

HoD/Coordinator

Examiner

1.0 Introduction

1.1 Background and Overview of Personal Finance

In an increasingly complex economic environment, the importance of financial literacy has grown to become a critical life skill. The ability for an individual to competently manage their personal finances—a discipline that encompasses budgeting, spending, saving, and investing—serves as the foundation for achieving long-term financial stability and personal life goals. Effective financial management provides the tools to navigate economic uncertainties, systematically build wealth, and ensure future security. This study transitions from theoretical concepts to practical application by delving into the real-world financial habits of a specific demographic, providing a snapshot of their behaviours and attitudes.

1.2 Purpose of the Study and Key Research Questions

The primary purpose of this research is to conduct a detailed analysis of the personal spending, saving, and investment behaviours of individuals residing in the Kalyan, Maharashtra area. By methodically collecting and examining data on their financial habits and attitudes, this study aims to identify prevailing trends, uncover common challenges, and understand the key factors that influence financial decision-making within this community.

To guide this investigation, the study is structured around the following core research questions:

- What are the most common methods used by individuals for tracking their personal expenses?
- What is the prevalence of regular saving habits among the surveyed population?
- How do demographic factors, with a particular focus on monthly income, influence saving and investment behaviour?
- What are the primary motivations that drive individuals to save, and what are the key influences on their investment decisions?

2.0 Methodology

2.1 Questionnaire Design

To gather comprehensive data, a structured questionnaire was designed and administered through a Google Form. The survey was logically divided into four distinct sections, each tailored to collect specific types of information:

1. **Demographics:** This section collected foundational data on age, gender, occupation, and monthly income to allow for segmentation and comparative analysis of different respondent groups.
2. **Spending Habits:** Questions in this part focused on identifying the specific methods respondents use for expense tracking and gauging their average monthly spending.
3. **Saving and Investment Habits:** This section was designed to assess the regularity of saving, the motivations behind it, current participation in investments, and preferences for different investment avenues.
4. **Financial Attitudes:** A 5-point Likert scale was employed to quantitatively measure respondents' level of agreement with statements about their financial confidence and future outlook, providing insight into their subjective financial well-being.

2.2 Data Collection and Sample Population

The data was collected using the online survey created with Google Forms. The survey link was actively distributed through social media platforms and messaging applications over a two-week period in late September 2025. This process yielded a total of **50 valid responses**. The study utilized a convenience sample, with respondents primarily residing in the **Kalyan, Maharashtra region**, which defines the geographical scope of the findings.

3.0 Results and Analysis

3.1 Demographic Profile of Respondents

The analysis of the 50 respondents revealed a diverse sample, with the following key characteristics:

- **Age Distribution:** The largest segment of respondents was the **18–24** age group, representing **27.5%** of the sample. This was followed by the 25–34 group at 21.6% and the 45+ group at 19.6%.
- **Occupation: Students** constituted the majority at **31.4%**. The unemployed were the next largest group at 17.6%.
- **Monthly Income:** The most common income bracket was **Less than ₹10,000**, accounting for **29.4%** of respondents.

3.2 Analysis of Spending and Expense Tracking Habits

The study found varied approaches to expense management. The most common method was **Using Apps** (e.g., Walnut, YNAB), adopted by **35.3%** of respondents. A nearly equal portion, **33.3%**, preferred **Manual** methods such as using a notebook or spreadsheet. A significant minority of **31.4%** reported that they **do not track their expenses at all**, indicating a lack of formal budgeting among almost a third of the sample.

3.3 Analysis of Saving and Investment Behaviour

A very strong positive correlation was observed between income levels and saving habits. The data shows a stark contrast: **95% of respondents earning above ₹50,000 reported a regular saving habit**. This is in sharp comparison to **only 40% of those earning less than ₹25,000** who save regularly.

Regarding investment, the study identified a significant gap between saving and investing. While many respondents save, **only 30% currently invest their money**. A larger group of **45%** reported that they are "Planning to start," suggesting a potential for future growth in investment activity but highlighting current hesitation.

3.4 Attitudes Towards Financial Management

Respondent confidence in their financial future was measured on a 5-point Likert scale, where 1 represented low confidence and 5 represented high confidence. the following results:

- **Mean Score:** 3.11
- **Median Score:** 3.0
- **Standard Deviation:** 1.39

The mean score, slightly above the neutral midpoint of 3, indicates a generally positive but not overwhelmingly strong level of financial confidence. The median of 3.0 confirms this, showing that half the respondents feel neutral or less confident. The standard deviation of 1.39 suggests a moderate spread in opinions, meaning that while the average is slightly positive, there is no strong consensus, and many individuals feel less secure.

4.0 Discussion of Key Findings

4.1 The "Aware but Manual" Spender

A key takeaway is that a high majority of respondents demonstrate financial awareness by actively tracking their expenses. However, the near-equal split between those using modern apps and those relying on manual methods suggests a significant portion of the population has not yet fully embraced digital finance tools. This preference for manual tracking could indicate barriers to entry for FinTech solutions, such as perceived complexity, a lack of trust in digital platforms, or simply a lack of awareness of the benefits these apps can offer.

4.2 The Strong Link Between Income and Savings

The data unequivocally demonstrates that income level is a primary determinant of saving behaviour. While this correlation is expected, the dramatic difference between high-income (95% save regularly) and low-income (40% save regularly) brackets is particularly telling. This disparity highlights the immense financial pressures faced by those with lower earnings, who may lack the necessary disposable income to establish a consistent saving habit, regardless of their financial literacy or intentions.

4.3 The Gap Between Saving and Investing

One of the most crucial insights is the significant "saver-investor gap". While a majority of respondents have established a habit of saving, only 30% have transitioned to investing their money to generate wealth. This indicates that while the principle of setting money aside is well understood, many are hesitant to take the next step. This hesitation likely stems from several factors, including a lack of investment knowledge, a fear of market risks, or the perception of having insufficient capital to begin investing.

5.0 Conclusion

5.1 Summary of Key Findings

- **Financial Awareness:** Most respondents are financially conscious and track their expenses, but a large number still depend on traditional manual methods.
- **Income-Driven Savings:** The practice of saving is common but is strongly dependent on the individual's income level.
- **The Saver-Investor Gap:** A significant gap exists wherein many individuals who save regularly are not yet investing their money.
- **Moderate Confidence:** Financial confidence is moderately positive on average but is not universal, reflecting underlying uncertainty among many respondents.

5.2 Recommendations

Based on the study's findings, the following actions are recommended:

- **For FinTech Companies:** There is a clear market opportunity for developing highly user-friendly financial applications. These apps should aim to simplify the transition from manual tracking to automated digital budgeting, making financial management more accessible.
- **For Financial Educators:** Educational workshops and online content should be specifically designed to bridge the gap between saving and investing. The focus should be on demystifying investment by explaining the basics of risk, the power of long-term growth, and compound interest in simple, clear terms.
- **For Individuals:** Individuals in lower-income brackets could greatly benefit from education on "micro-saving" and "micro-investing" strategies. These approaches can help in building foundational financial habits and discipline, even with limited funds.

5.3 Limitations and Future Research

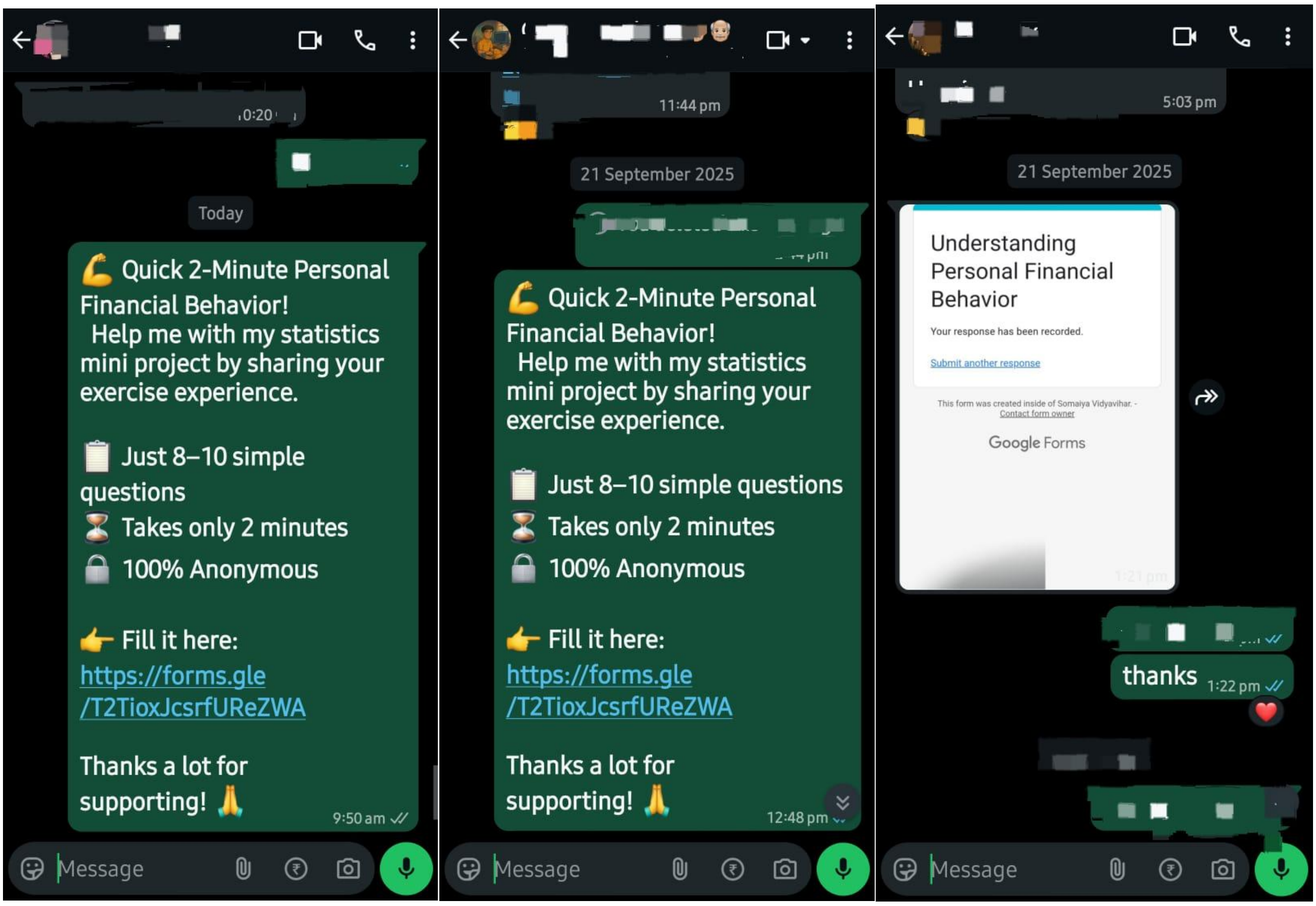
The conclusions of this study should be considered in light of its limitations. The research was conducted with a small convenience sample from a specific geographic area (Kalyan, Maharashtra), which limits the generalizability of the findings to a wider population.

For future research, it is recommended to pursue two main avenues:

1. Conduct a similar study with a larger, more diverse, and randomized sample to improve statistical validity and generalizability.
2. Employ a longitudinal study design to track how the financial habits and attitudes of a cohort evolve over time, providing deeper insights into financial decision-making journeys.

6.0 References

6.1 Evidences



CODE & OUTPUT (link)⇏

```
# Setup: Loading and Preparing Your
Data
#

# Step 1: Install and load necessary packages
# install.packages("readxl")
# install.packages("tidyverse")
library(readxl)
library(tidyverse)

# Step 2: Load your Excel file
my_data <- read_excel("C:\\Users\\mural\\Downloads\\31011224039 (Responses).xlsx")


# Step 3: Rename columns for easy access. This is a crucial step.
# Match the text in quotes to the exact column headers from your Excel file.
my_data <- my_data %>%
  rename(
    Age = `Age`,
    Gender = `Gender`,
    Occupation = `Occupation`,
    MonthlyIncome = `Monthly Income (Approximate)`,
    ExpenseTracking = `How do you usually track your expenses?`,
    SavingHabit = `Do you have a regular habit of saving money?`,
    AvgSavings = `How much of your monthly income do you save on average?`,
    InvestsMoney = `Do you invest your money?`,
    # For the Likert scale, you'll have multiple columns. Name them systematically.
    # Let's assume the first statement was about financial confidence.
    FinancialConfidence = `Please indicate your level of agreement with the
following statements:`
  )


# Step 4: Set an order for categorical data
# This ensures your charts (e.g., for income) are in a logical order, not
alphabetical.
my_data$MonthlyIncome <- factor(my_data$MonthlyIncome, levels = c(
  "Less than ₹10,000", "₹10,000-₹25,000", "₹25,001-₹50,000",
  "₹50,001-₹1,00,000", "Above ₹1,00,000"
))
my_data$FinancialConfidence <- as.numeric(my_data$FinancialConfidence)
```

```r
# Frequency Distribution and
Percentages
#

# --- Analysis of Expense Tracking Methods ---
tracking_summary <- my_data %>%
  count(ExpenseTracking, sort = TRUE) %>%
  mutate(Percentage = round((n / sum(n)) * 100, 1))

print("How Respondents Track Expenses:")
print(tracking_summary)
```

```
[1] "How Respondents Track Expenses:"
> print(tracking_summary)
# A tibble: 3 × 3
  ExpenseTracking                      n Percentage
  <chr>                            <int>      <dbl>
1 Using Apps (e.g., Walnut, YNAB)     18       35.3
2 Manually (Notebook/Spreadsheet)     17       33.3
3 I don't track my expenses           16       31.4
>
```

```r
# --- Analysis of Age Group Distribution ---
age_summary <- my_data %>%
  # Make sure the Age column is an ordered factor for logical sorting
  mutate(Age = factor(Age, levels = c("Under 18", "18-24", "25-34", "35-44",
"45+"))) %>%
  count(Age) %>%
  # Calculate the percentage for each group
  mutate(Percentage = round((n / sum(n)) * 100, 1))

print("Demographic Analysis: Age Group Distribution")
print(age_summary)
```

```
> print("Demographic Analysis: Age Group Distribut
[1] "Demographic Analysis: Age Group Distribution"
> print(age_summary)
# A tibble: 5 × 3
  Age          n Percentage
  <fct>    <int>      <dbl>
1 Under 18     8       15.7
2 18-24       14       27.5
3 25-34       11       21.6
4 35-44        8       15.7
5 45+         10       19.6
```

```r
# --- Analysis of Occupation Distribution ---
occupation_summary <- my_data %>%
  # Count each occupation, sorting by the most frequent
  count(Occupation, sort = TRUE) %>%
  mutate(Percentage = round((n / sum(n)) * 100, 1))

print("Demographic Analysis: Occupation Distribution")
print(occupation_summary)
```

```
[1] "Demographic Analysis: Occupation Distribution"
> print(occupation_summary)
# A tibble: 5 × 3
  Occupation            n Percentage
  <chr>             <int>      <dbl>
1 Student              16       31.4
2 Retired               9       17.6
3 Salaried Employee     9       17.6
4 Self-Employed         9       17.6
5 Unemployed            8       15.7
```

```r
# --- Analysis of Monthly Income Distribution ---
# The 'MonthlyIncome' column was already converted to an ordered factor in a previous step,
# which is essential for this analysis to be sorted correctly.

income_summary <- my_data %>%
  count(MonthlyIncome) %>%
  mutate(Percentage = round((n / sum(n)) * 100, 1))

print("Demographic Analysis: Monthly Income Distribution")
print(income_summary)
```

```
[1] "Demographic Analysis: Monthly Income Distribution"
> print(income_summary)
# A tibble: 5 × 3
  MonthlyIncome           n Percentage
  <fct>               <int>      <dbl>
1 Less than ₹10,000      15       29.4
2 ₹10,000-₹25,000         9       17.6
3 ₹25,001-₹50,000         9       17.6
4 ₹50,001-₹1,00,000       9       17.6
5 Above ₹1,00,000         9       17.6
>
```

```r
# --- Analysis of Saving Habits ---
saving_habit_summary <- my_data %>%
  count(SavingHabit, sort = TRUE) %>%
  mutate(Percentage = round((n / sum(n)) * 100, 1))

print("Regularity of Saving Habits:")
print(saving_habit_summary)
```

```
[1] "Regularity of Saving Habits:"
> print(saving_habit_summary)
# A tibble: 3 × 3
  SavingHabit     n Percentage
  <chr>       <int>      <dbl>
1 Yes            21       41.2
2 No             15       29.4
3 Sometimes      15       29.4
```

```r
# --- Analysis of Likert Scale Data (Financial Confidence) ---
# First, ensure the column is numeric (it will be read as 1, 2, 3, 4, 5)
my_data$FinancialConfidence <- as.numeric(my_data$FinancialConfidence)

confidence_stats <- my_data %>%
  summarise(
    AverageScore = mean(FinancialConfidence, na.rm = TRUE),
    MedianScore = median(FinancialConfidence, na.rm = TRUE)
  )

print("Financial Confidence Score (1-5 scale):")
print(confidence_stats)
```

```
[1] "Financial Confidence Score (1-5 scale):"
> print(confidence_stats)
# A tibble: 1 × 2
  AverageScore MedianScore
         <dbl>       <dbl>
1         3.12           3
>
```

```r
# Measures of Central Tendency &
Dispersion
#

# Central Tendency
mean_confidence <- mean(my_data$FinancialConfidence, na.rm = TRUE)
median_confidence <- median(my_data$FinancialConfidence, na.rm = TRUE)

# Dispersion
range_confidence <- range(my_data$FinancialConfidence, na.rm = TRUE)
sd_confidence <- sd(my_data$FinancialConfidence, na.rm = TRUE)
var_confidence <- var(my_data$FinancialConfidence, na.rm = TRUE)

# Print the results
cat("Mean Financial Confidence Score:", mean_confidence, "\n")
cat("Median Financial Confidence Score:", median_confidence, "\n")
cat("Score Range:", range_confidence[1], "-", range_confidence[2], "\n")
cat("Standard Deviation:", sd_confidence, "\n")
```

```
Mean Financial Confidence Score: 3.117647
> cat("Median Financial Confidence Score:
Median Financial Confidence Score: 3
> cat("Score Range:", range_confidence[1]
Score Range: 1 - 5
> cat("Standard Deviation:", sd_confidenc
Standard Deviation: 1.394949
>
```

```r
# --- Finding the Mode for a Categorical Variable ---
# R doesn't have a built-in mode function, so we find the most frequent item
mode_tracking_method <- my_data %>%
  count(ExpenseTracking, sort = TRUE) %>%
  # The 'pull()' function extracts the first value from the column
  pull(ExpenseTracking) %>%
  first()

cat("Mode (Most Common) Expense Tracking Method:", mode_tracking_method, "\n")
```

```
>
> cat("Mode (Most Common) Expense Tracking Method:", mode_tracking_method, "\n")
Mode (Most Common) Expense Tracking Method: Using Apps (e.g., Walnut, YNAB)
```

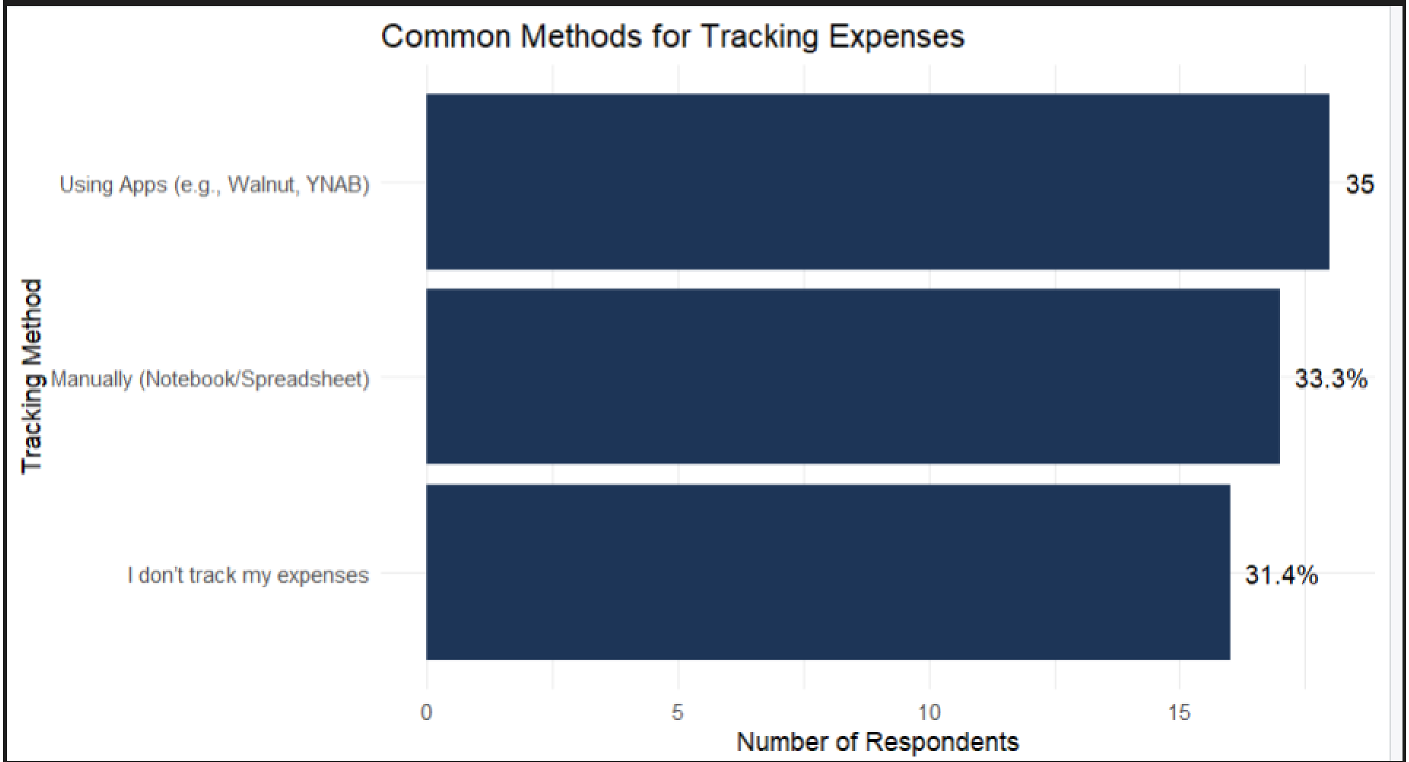
```r
## Insights from Demographic
Differences
#

# --- Cross-Tabulation: Monthly Income vs. Saving Habit ---
income_vs_saving <- my_data %>%
  # Filter out any rows where the data might be empty
  filter(!is.na(MonthlyIncome) & !is.na(SavingHabit)) %>%
  count(MonthlyIncome, SavingHabit) %>%
  group_by(MonthlyIncome) %>%
  mutate(Percentage = round((n / sum(n)) * 100, 1)) %>%
  # ungroup() is good practice after a group_by operation
  ungroup()

print("Comparison of Saving Habits Across Income Brackets:")
print(income_vs_saving)
```

```
[1] "Comparison of Saving Habits Across Income Brackets:"
> print(income_vs_saving)
# A tibble: 15 × 4
   MonthlyIncome      SavingHabit      n Percentage
   <fct>              <chr>        <int>      <dbl>
 1 Less than ₹10,000 No               4       26.7
 2 Less than ₹10,000 Sometimes        3       20
 3 Less than ₹10,000 Yes              8       53.3
 4 ₹10,000-₹25,000    No               4       44.4
 5 ₹10,000-₹25,000    Sometimes        3       33.3
 6 ₹10,000-₹25,000    Yes              2       22.2
 7 ₹25,001-₹50,000    No               3       33.3
 8 ₹25,001-₹50,000    Sometimes        2       22.2
 9 ₹25,001-₹50,000    Yes              4       44.4
10 ₹50,001-₹1,00,000 No               2       22.2
11 ₹50,001-₹1,00,000 Sometimes        4       44.4
12 ₹50,001-₹1,00,000 Yes              3       33.3
13 Above ₹1,00,000    No               2       22.2
14 Above ₹1,00,000    Sometimes        3       33.3
15 Above ₹1,00,000    Yes              4       44.4
> 
```

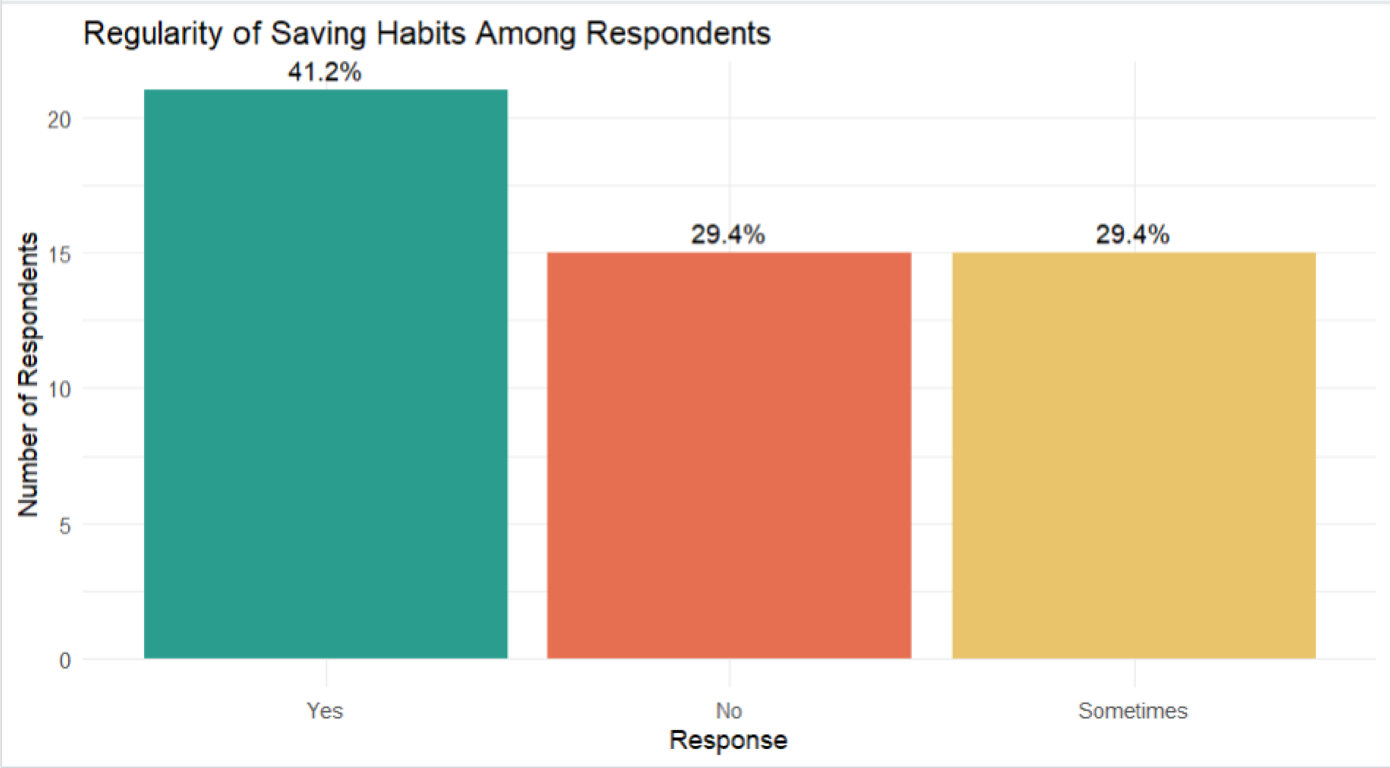
```r
# Create a summary table that counts the combinations of saving and investment habits.
saving_vs_investment <- my_data %>%
  # Ensure we only use rows with complete data for both columns
  filter(!is.na(SavingHabit) & !is.na(InvestsMoney)) %>%
  # Count the occurrences of each pair
  count(SavingHabit, InvestsMoney) %>%
  # Group by the saving habit to calculate percentages within that group
  group_by(SavingHabit) %>%
  # Calculate the percentage for each investment status within each saving habit
  mutate(Percentage = round((n / sum(n)) * 100, 1)) %>%
  ungroup() # Ungroup for good practice

print("Summary of Saving vs. Investment Habits:")
print(saving_vs_investment)
```

```
[1] "Summary of Saving vs. Investment Habits:"
> print(saving_vs_investment)
# A tibble: 6 × 4
  SavingHabit InvestsMoney          n Percentage
  <chr>       <chr>             <int>      <dbl>
1 No          No                   13       86.7
2 No          Yes                   2       13.3
3 Sometimes   Planning to start    15      100  
4 Yes         No                    3       14.3
5 Yes         Planning to start     2        9.5
6 Yes         Yes                  16       76.2
> 
```

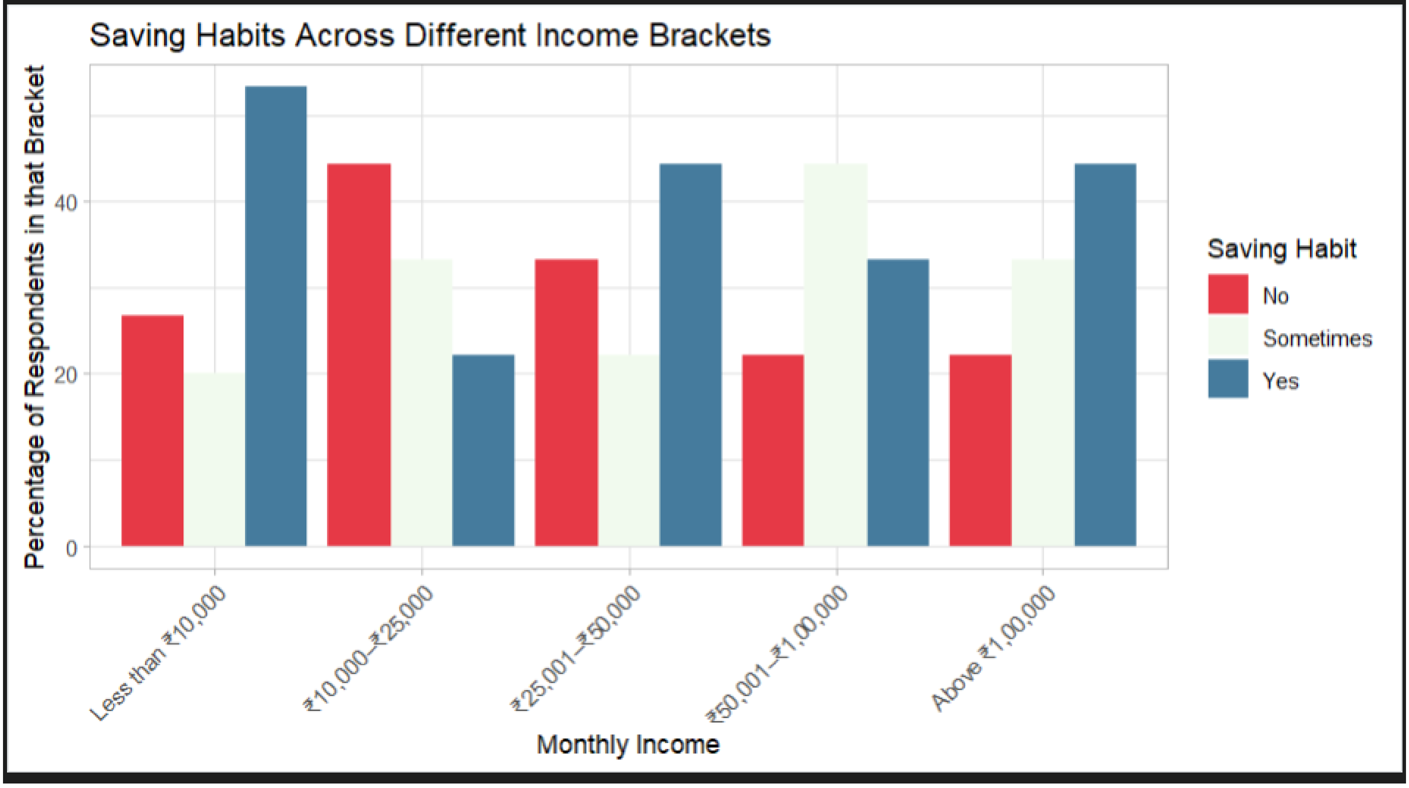
```
## Visualization for Your
Report
#

ggplot(tracking_summary, aes(x = reorder(ExpenseTracking, n), y = n)) +
  geom_bar(stat = "identity", fill = "#1d3557") +
  geom_text(aes(label = paste0(Percentage, "%")), hjust = -0.2) + # Add percentage
labels
  labs(
    title = "Common Methods for Tracking Expenses",
    x = "Tracking Method",
    y = "Number of Respondents"
  ) +
  coord_flip() + # Flip coordinates for better readability of labels
  theme_minimal()
```



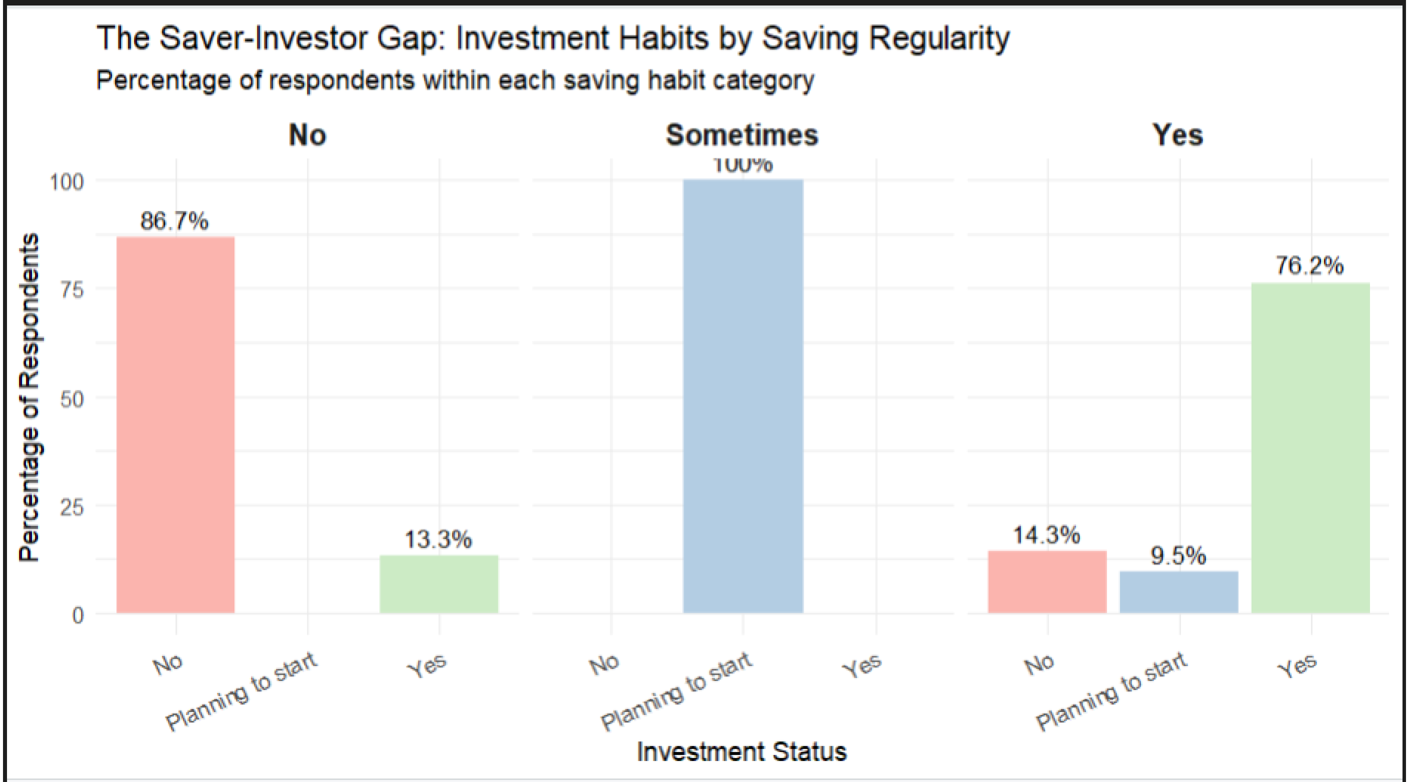
```r
ggplot(saving_habit_summary, aes(x = reorder(SavingHabit, -n), y = n, fill =
SavingHabit)) +
  geom_bar(stat = "identity", show.legend = FALSE) +
  geom_text(aes(label = paste0(Percentage, "%")), vjust = -0.5) + # Add percentage
labels
  labs(
    title = "Regularity of Saving Habits Among Respondents",
    x = "Response",
    y = "Number of Respondents"
  ) +
  scale_fill_manual(values = c("Yes" = "#2a9d8f", "No" = "#e76f51", "Sometimes" =
"#e9c46a")) +
  theme_minimal()
```



```r
ggplot(income_vs_saving, aes(x = MonthlyIncome, y = Percentage, fill =
SavingHabit)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Saving Habits Across Different Income Brackets",
    x = "Monthly Income",
    y = "Percentage of Respondents in that Bracket",
    fill = "Saving Habit"
  ) +
  scale_fill_manual(values = c("Yes" = "#457b9d", "No" = "#e63946", "Sometimes" =
"#f1faee")) +
  theme_light() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```r
# This chart creates separate plots for each saving habit for a clear comparison.
ggplot(saving_vs_investment, aes(x = InvestsMoney, y = Percentage, fill =
InvestsMoney)) +
  geom_bar(stat = "identity", show.legend = FALSE) +
  # Add the percentage labels on top of the bars
  geom_text(aes(label = paste0(Percentage, "%")), vjust = -0.5, size = 3.5) +
  # Create separate charts for "Yes", "No", and "Sometimes"
  facet_wrap(~ SavingHabit) +
  labs(
    title = "The Saver-Investor Gap: Investment Habits by Saving Regularity",
    subtitle = "Percentage of respondents within each saving habit category",
    x = "Investment Status",
    y = "Percentage of Respondents"
  ) +
  scale_y_continuous(limits = c(0, 100)) + # Set y-axis from 0 to 100
  scale_fill_brewer(palette = "Pastel1") + # Use a nice color palette
  theme_minimal() +
  theme(
    strip.text = element_text(size = 12, face = "bold"), # Style the facet titles
    axis.text.x = element_text(angle = 25, hjust = 1)
  )
```



```r
ggplot(tracking_summary, aes(x = "", y = Percentage, fill = ExpenseTracking)) +
  # Create a stacked bar chart
  geom_bar(stat = "identity", width = 1, color = "white") +
  # Convert the bar chart to a pie chart
  coord_polar("y", start = 0) +
  # Add the percentage labels to the middle of each slice
  geom_text(aes(label = paste0(Percentage, "%")),
            position = position_stack(vjust = 0.5),
            color = "black",
            size = 5) +
  # Use a nice color palette
  scale_fill_brewer(palette = "Pastel1") +
  # Use a clean theme with no axis labels or gridlines
  theme_void() +
  # Add a title and a better legend title
  labs(
    title = "How Respondents Track Their Expenses",
    fill = "Tracking Method"
  )
```